CS2800 - Final Project

Fall 2026 | Interactive E-Commerce Website

Total Points: 120
Project Type: Individual
Technologies: HTML, CSS, JavaScript, and `localStorage`

Project Overview

For the final project, you will design and develop an e-commerce website for a store of your choice. Your website should allow customers to browse products, manage a shopping cart, complete a simulated purchase, and view previous orders.

You will use HTML and CSS to build the website and JavaScript to implement its functionality. Since this project does not require a database or server, you will use `localStorage` to save information between visits.

Your project should demonstrate the web development skills covered throughout the semester.

1. Website Requirements

Your website must include the following features.

Product Catalog

- At least 12 products across 3 or more categories.
- Each product must have a name, image, description, price, and available quantity.
- Products must be displayed using JavaScript.
- Customers must be able to search, filter by category, and sort products by price.
- Each product must have a detail view.

Shopping Cart

- Add and remove products.
- Change the quantity of products in the cart.
- Automatically calculate subtotals and the cart total.
- Prevent customers from adding more items than are in stock.
- Save the cart using `localStorage` so it remains available after refreshing the page.

Checkout

Create a checkout form that collects the customer's name, email, and shipping address.

The checkout must:

- Validate required information before submitting an order.
- Offer standard shipping ($5) and express shipping ($15).
- Provide free standard shipping on orders of $100 or more.
- Display the final order total.
- Generate an order confirmation after a successful purchase.

No real payment processing is required.

Inventory and Order History

- Each product must have an initial stock quantity.
- Completing a purchase must reduce the available stock.
- Inventory changes must be saved using `localStorage`.
- Customers must be able to view their previous orders, including purchased items, quantities, and totals.
- Orders must remain available after refreshing the page.
- Include an option to reset the store's demonstration data.

Your application should handle common problems, such as empty carts, invalid quantities, out-of-stock products, and missing saved data.

2. Design and Technical Requirements

Your website must:

- Have at least five pages or equivalent views: Home, Product Details, Cart, Checkout, and Order History.
- Use semantic HTML and external CSS and JavaScript files.
- Have consistent navigation and styling.
- Work on both desktop and mobile screens.
- Include properly labeled forms and meaningful image descriptions.
- Use JavaScript arrays, objects, functions, events, and DOM manipulation.
- Use `localStorage` for the cart, inventory, and order history.

You may not use JavaScript frameworks, CSS frameworks, back-end services, databases, or website builders.

3. Individual Requirements

In addition to the requirements above, each student will receive **two additional features** assigned by the instructor.

Possible features include:

Feature	**Description**
Discount system	Apply a discount based on specified conditions
Wishlist	Allow customers to save favorite products
Low-stock warning	Display a warning when inventory falls below a set quantity
Order filtering	Filter previous orders based on certain criteria
Promotional code	Apply a discount when a valid code is entered
Recently viewed	Display products the customer previously viewed

The instructor will assign the features and any specific conditions after the first checkpoint.

These requirements are part of the project grade.

4. Project Checkpoints

The project will be completed in stages. Dates will be announced in class.

Checkpoint	**Expected progress**
1. Planning and Design	Store concept, wireframes, product information, and initial HTML structure
2. Products and Cart	Working product catalog, search/filter features, shopping cart, and `localStorage`
3. Checkout and Orders	Checkout, inventory updates, order history, and individual requirements

Checkpoint	Expected progress
Final Submission	Completed website, GitHub repository, and short report

Students must use GitHub throughout development. Each checkpoint should be reflected in the repository's commit history.

5. Final Submission

Submit your completed project through GitHub.

Your repository must include:

- All HTML, CSS, JavaScript, and required images.
- A `README.md` explaining how to run the website.
- A short report (approximately 400–600 words) describing your design, JavaScript implementation, use of `localStorage`, and two problems you encountered during development.
- A record of testing the major website features.

The website should run locally without requiring a paid service or external database.

6. Final Presentation and Oral Examination

Each student will complete an individual demonstration and oral examination.

Duration: Approximately 10–12 minutes.

During the examination, you will:

1. Demonstrate the major features of your website.
2. Explain selected sections of your HTML, CSS, and JavaScript.
3. Explain how your shopping cart, inventory, and orders use `localStorage`.
4. Answer questions about your programming decisions.

You should be familiar with all code submitted as part of your project.

Live Programming Exercise

You will also be asked to make a small change to your website during the examination.

For example, you may be asked to modify a shipping rule, fix a cart calculation, add a validation condition, or change how a product is displayed.

The exercise will take approximately 5–8 minutes and will involve concepts covered in class.

You will be graded on your understanding, approach, implementation, and ability to explain your changes.

7. Academic Integrity and AI Policy

This is an individual programming assignment.

The use of generative AI tools, including ChatGPT, GitHub Copilot, Claude, and Gemini, is not permitted for this project. This includes generating code, debugging, writing documentation, creating designs, or completing any other project component.

You may use course materials, class notes, official documentation, MDN Web Docs, and standard browser developer tools.

You may discuss general course concepts with classmates, but all submitted work must be your own.

Students must be able to explain and modify their code independently during the final examination.

Academic integrity concerns will be addressed according to university policy.

8. Grading Rubric

Component	**Points**
Product catalog, search, filtering, and sorting	10
Shopping cart functionality	10
Checkout and order processing	10
Inventory, order history, and `localStorage`	15
HTML, CSS, responsive design, and code quality	20
Individual assigned features	10

Component	**Points**
Live programming exercise	10
Documentation	15
Checkpoints	10
Oral examination/Presentation	10
Total	**120**

Features will be graded based on whether they work correctly, including when users enter invalid information or perform unexpected actions.

A working website is only part of the project. You must also demonstrate that you understand how it works and can make changes to your code.

Final Reminder: Your website does not need to look like a commercial online store. Focus on creating a functional, organized, and usable application that demonstrates what you have learned in CS2800.